5. Visualization of various massive dataset - Finance - Healthcare-Census-Geospatial.

a) Finance:

Step 1:-

```
import pandas as pd
import numpy as np
from datetime import datetime, timedelta
# Generate synthetic data
np.random.seed(42)
date_today = datetime.now()
days = pd.date_range(date_today, date_today + timedelta(29), freq='D')
data = {'Date': days, 'StockPrice': np.random.randint(100, 200, size=(30,))}
df = pd.DataFrame(data)
# Save the dataset to a CSV file
df.to_csv('finance_dataset.csv',index=False)
```

Step 2:

```
import pandas as pd
import matplotlib.pyplot as plt
# Read the dataset from the CSV file
df = pd.read_csv('finance_dataset.csv')
# Convert the 'Date' column to datetime format
df['Date'] = pd.to_datetime(df['Date'])

# Plotting the financial data
plt.figure(figsize=(10, 6))
plt.plot(df['Date'], df['StockPrice'], marker='o', linestyle='-', color='b')

# Adding labels and title
plt.title('Stock Prices Over Time')
plt.xlabel('Date')
plt.ylabel('Stock Price')

# Display the plot
plt.grid(True)
plt.show()
```

Expected Output:-



b) Healthcare:
Dataset: https://drive.google.com/file/d/1YQKfblYW9L4p-JOn2XRibr7VMa3STh86/view?usp=sharing

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load your healthcare dataset (replace 'path/to/your/healthcare_dataset.csv' with the actual path)
healthcare_data = pd.read_csv('healthcare_dataset.csv')

# Display basic information about the dataset
print("Dataset Overview:")
print(healthcare_data.info())

# Display the first few rows of the dataset
print("\nFirst Few Rows of the Dataset:")
print(healthcare_data.head())

# Descriptive statistics of numerical columns
print("\nDescriptive Statistics:")
print(healthcare_data.describe())

# Handling missing data
print("\nHandling Missing Data:")
print(healthcare_data.isnull().sum())

# Data visualization

# Example: Bar plot for patient gender distribution
plt.figure(figsize=(8, 5))
sns.countplot(x='Gender', data=healthcare_data)
plt.title('Patient Gender Distribution')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.show()

# Example: Box plot for age distribution by gender
plt.figure(figsize=(10, 6))
sns.boxplot(x='Gender', y='Age', data=healthcare_data)
plt.title('Age Distribution by Gender')
plt.xlabel('Gender')
plt.ylabel('Age')
plt.show()

# Example: Heatmap for correlation between numerical variables
plt.figure(figsize=(10, 8))
sns.heatmap(healthcare_data.corr(), annot=True, cmap='coolwarm', fmt='.2f', linewidths=0.5)
plt.title('Correlation Heatmap')
plt.show()
```

Expected Output:

```
Dataset Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 15 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Name                10000 non-null  object
 1   Age                 10000 non-null  int64
 2   Gender              10000 non-null  object
 3   Blood Type          10000 non-null  object
 4   Medical Condition   10000 non-null  object
 5   Date of Admission   10000 non-null  object
 6   Doctor              10000 non-null  object
 7   Hospital            10000 non-null  object
 8   Insurance Provider  10000 non-null  object
 9   Billing Amount      10000 non-null  float64
 10  Room Number         10000 non-null  int64
 11  Admission Type      10000 non-null  object
 12  Discharge Date      10000 non-null  object
 13  Medication          10000 non-null  object
 14  Test Results        10000 non-null  object
dtypes: float64(1), int64(2), object(12)
memory usage: 1.1+ MB
None

First Few Rows of the Dataset:
                Name  Age  Gender Blood Type Medical Condition  \
0      Tiffany Ramirez   81  Female         O-          Diabetes
1         Ruben Burns   35    Male         O+           Asthma
2           Chad Byrd   61    Male         B-          Obesity
3    Antonio Frederick   49    Male         B-           Asthma
4  Mrs. Brandy Flowers   51    Male         O-        Arthritis

  Date of Admission          Doctor                Hospital  \
0        2022-11-17  Patrick Parker          Wallace-Hamilton
1        2023-06-01   Diane Jackson  Burke, Griffin and Cooper
2        2019-01-09      Paul Baker               Walton LLC
3        2020-05-02  Brian Chandler               Garcia Ltd
4        2021-07-09  Dustin Griffin    Jones, Brown and Murray

  Insurance Provider  Billing Amount  Room Number Admission Type  \
0          Medicare    37490.983364          146       Elective
1   UnitedHealthcare    47304.064845          404      Emergency
2          Medicare    36874.896997          292      Emergency
3          Medicare    23303.322092          480         Urgent
4   UnitedHealthcare    18086.344184          477         Urgent

  Discharge Date   Medication  Test Results
0     2022-12-01      Aspirin  Inconclusive
```

1 2023-06-15 Lipitor Normal
2 2019-02-08 Lipitor Normal
3 2020-05-03 Penicillin Abnormal
4 2021-08-02 Paracetamol Normal

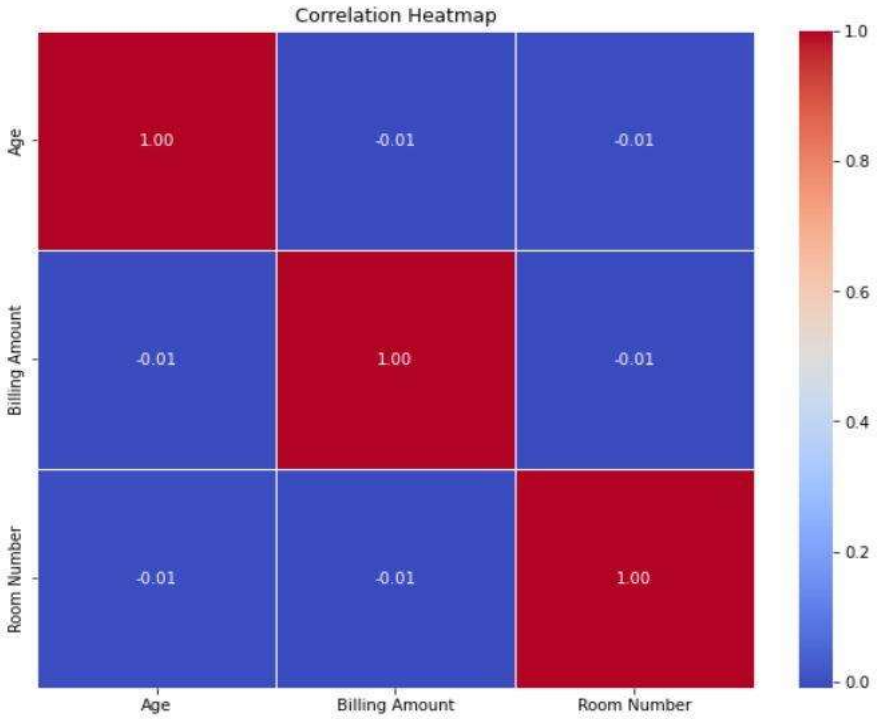
Descriptive Statistics:

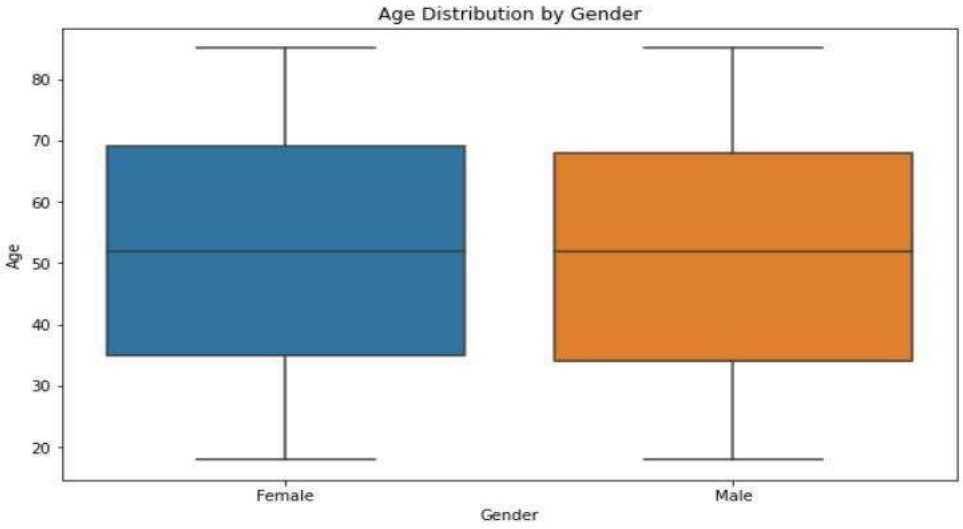
	Age	Billing Amount	Room Number
count	10000.000000	10000.000000	10000.000000
mean	51.452200	25516.806778	300.082000
std	19.588974	14067.292709	115.806027
min	18.000000	1000.180837	101.000000
25%	35.000000	13506.523967	199.000000
50%	52.000000	25258.112566	299.000000
75%	68.000000	37733.913727	400.000000
max	85.000000	49995.902283	500.000000

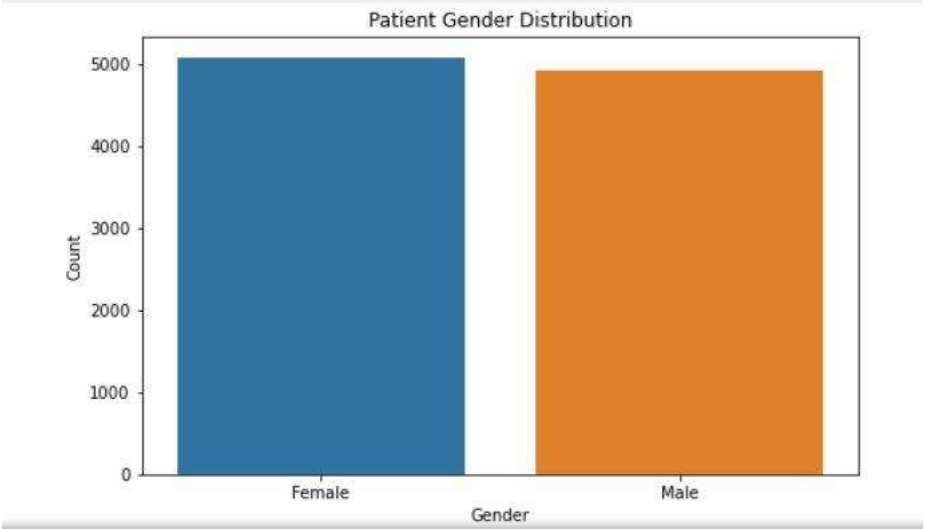
Handling Missing Data:

Name	0
Age	0
Gender	0
Blood Type	0
Medical Condition	0
Date of Admission	0
Doctor	0
Hospital	0
Insurance Provider	0
Billing Amount	0
Room Number	0
Admission Type	0
Discharge Date	0
Medication	0
Test Results	0

dtype: int64


Correlation Heatmap
Age
Billing Amount
Room Number
1.00
-0.01
-0.01
-0.01
1.00
-0.01
-0.01
-0.01
1.00
Age
Billing Amount
Room Number
1.0
0.8
0.6
0.4
0.2
0.0


Age Distribution by Gender
Age
80
70
60
50
40
30
20
Female
Male
Gender


Patient Gender Distribution
Count
5000
4000
3000
2000
1000
0
Female
Male
Gender

c) Census

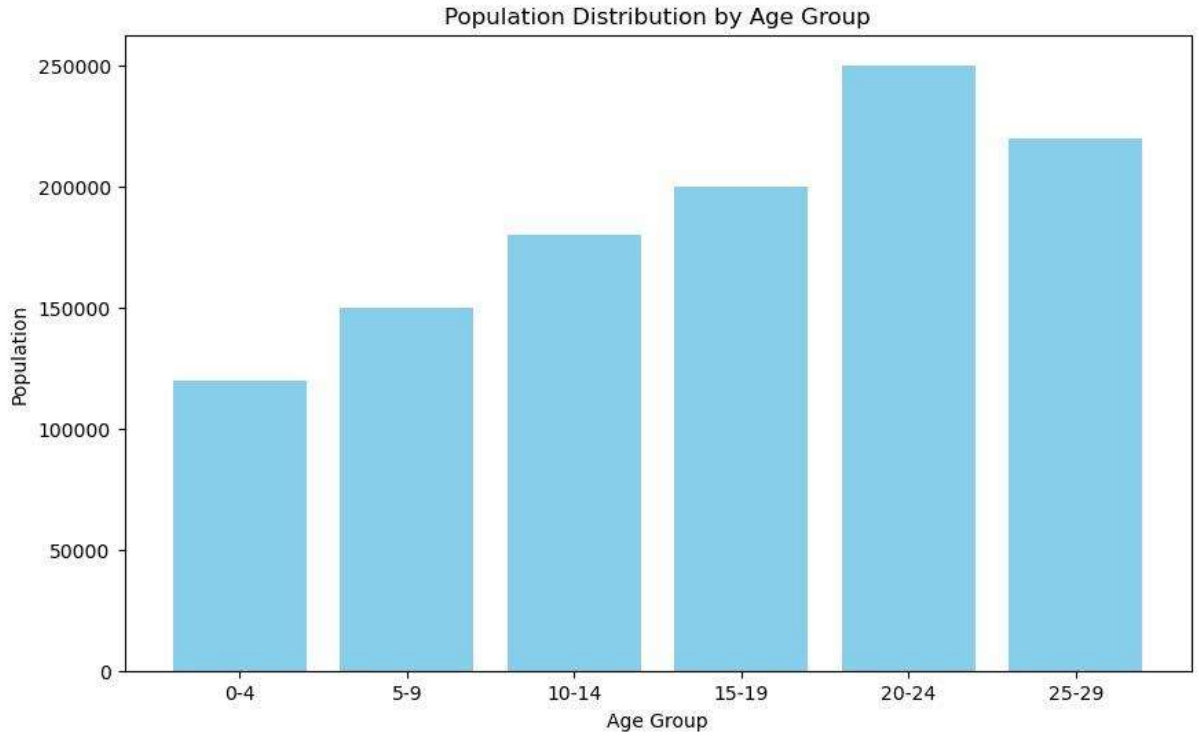
```
import matplotlib.pyplot as plt
import pandas as pd

# Assuming you have a pandas DataFrame with columns 'Age' and 'Population'
# Replace this with your actual data loading mechanism
# Example data creation:
data = {
    'Age': ['0-4', '5-9', '10-14', '15-19', '20-24', '25-29'],
    'Population': [120000, 150000, 180000, 200000, 250000, 220000]
}

df = pd.DataFrame(data)

# Plotting the data
plt.figure(figsize=(10, 6))
plt.bar(df['Age'], df['Population'], color='skyblue')
plt.xlabel('Age Group')
plt.ylabel('Population')
plt.title('Population Distribution by Age Group')
plt.show()
```

Expected Output:



d) Geopatial

Step 1:-

```
!pip install geopandas folium
```

Step 2:-

```
import geopandas as gpd
import folium
# Create a GeoDataFrame with example geospatial data
data = {
    'City': ['City A', 'City B', 'City C', 'City D'],
    'Country': ['Country 1', 'Country 2', 'Country 1', 'Country 3'],
    'Latitude': [34.0522, 40.7128, 41.8781, 37.7749],
    'Longitude': [-118.2437, -74.0060, -87.6298, -122.4194],
}
geometry = gpd.points_from_xy(data['Longitude'], data['Latitude'])
geo_df = gpd.GeoDataFrame(data, geometry=geometry)
# Display the GeoDataFrame
print("GeoDataFrame Information:")
print(geo_df)
# Plot the GeoDataFrame
geo_df.plot(marker='o', color='red', markersize=50, figsize=(10, 6))
plt.title('Geospatial Data Visualization')
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.show()
# Create an interactive map using folium
my_map = folium.Map(location=[geo_df['Latitude'].mean(), geo_df['Longitude'].mean()],
zoom_start=4)
# Add markers to the map
for index, row in geo_df.iterrows():
    folium.Marker(
        location=[row['Latitude'], row['Longitude']],
        popup=f"{row['City']}, {row['Country']}",
        icon=folium.Icon(color='blue')
    ).add_to(my_map)
# Save the map as an HTML file (optional)
my_map.save('geospatial_map.html')
# Display the map
my_map
```

Expected Output:-

```
GeoDataFrame Information:
     City    Country  Latitude  Longitude                    geometry
0  City A  Country 1   34.0522  -118.2437  POINT (-118.24370 34.05220)
1  City B  Country 2   40.7128   -74.0060   POINT (-74.00600 40.71280)
2  City C  Country 1   41.8781   -87.6298   POINT (-87.62980 41.87810)
3  City D  Country 3   37.7749  -122.4194  POINT (-122.41940 37.77490)
```

